

Ex. 1.1.2 Sauer 3

$$a) \quad x^5 + x = 1$$

-1 -1

since $f(0) < 0$ and $f(1) > 0$
there is a root in

$$(0, 1)$$

$$\Rightarrow f(x) = x^5 + x - 1$$

$$\Rightarrow f(0) = 0^5 + 0 - 1$$

$$\Rightarrow f(0) = -1$$

$$\Rightarrow f(1) = 1^5 + 1 - 1$$

$$\Rightarrow f(1) = 1$$

$$b) \quad \sin(x) = 6x + 5$$

-6x -5 -6x -5

root in interval

$$(-1, 0)$$

$$\Rightarrow f(x) = \sin(x) - 6x - 5$$

$$\Rightarrow f(-1) = \sin(-1) - 6(-1) - 5$$

$$\Rightarrow f(-1) = \sin(-1) + 1 \approx 0.159$$

$$\Rightarrow f(0) = \sin(0) - 6(0) - 5$$

$$\Rightarrow f(0) = -5$$

$$c) \quad \ln(x) + x^2 = 3$$

-3 -3

$$\Rightarrow f(x) = \ln(x) + x^2 - 3$$

$$\Rightarrow f(1) = \ln(1) + 1^2 - 3$$

$$\Rightarrow f(1) = -2$$

$$\Rightarrow f(2) = \ln(2) + 2^2 - 3$$

$$\Rightarrow f(2) = \ln(2) + 1 \approx 1.693$$

root at interval

$$(1, 2)$$

Ex. 1.1.4, Sauer 73

$$a) x^5 + x = 1$$

our interval was $(0, 1)$

$$f(x) = x^5 + x - 1$$

$$\text{Mid point} = \frac{0+1}{2} = \frac{1}{2}$$

we go from $(0, \frac{1}{2})$

$$\Rightarrow f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^5 + \frac{1}{2} - 1$$

$$\Rightarrow f\left(\frac{1}{2}\right) = -\frac{15}{32}$$

so the root is $(-\frac{15}{32}, 0)$

$$\text{mid point 2} = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4}$$

$$f\left(\frac{3}{4}\right) = \left(\frac{3}{4}\right)^5 + \frac{3}{4} - 1$$

$$\Rightarrow f\left(\frac{3}{4}\right) = -\frac{13}{1024} \quad \text{so root is } \left(\frac{3}{4}, 1\right)$$

Interval has length $1 - \frac{3}{4} = \frac{1}{4}$ so it's midpoint

is

$$x = \frac{7}{8}, \text{ root in } \left(\frac{3}{4}, 1\right)$$

b) $\sin(x) = 6x + 5$

$$f(x) = \sin(x) - 6x - 5$$

our interval was

$$(-1, 0)$$

$$\text{Midpoint} = \frac{-1 + 0}{2} = -\frac{1}{2}$$

$$f\left(-\frac{1}{2}\right) = \sin\left(-\frac{1}{2}\right) - 6\left(-\frac{1}{2}\right) - 5$$

$$\Rightarrow \sin\left(-\frac{1}{2}\right) - 2$$

$$\Rightarrow f\left(-\frac{1}{2}\right) \approx -2.4794$$

root is in $\left(-1, -\frac{1}{2}\right)$

$$\text{Midpoint 2} = \frac{-1 + \left(-\frac{1}{2}\right)}{2} = -\frac{3}{4}$$

$$f\left(-\frac{3}{4}\right) = \sin\left(-\frac{3}{4}\right) - 6\left(-\frac{3}{4}\right) - 5$$

$$f\left(-\frac{3}{4}\right) \approx -1.1816$$

root is $\left(-1, -\frac{3}{4}\right)$

interval has length $-\frac{3}{4} - (-1) = \frac{1}{4}$

$$\text{Midpoint} = \frac{-1 + \left(-\frac{3}{4}\right)}{2} = -\frac{7}{8} \approx -0.875$$

$$x = -\frac{7}{8}, \text{ root in } \left(-1, -\frac{3}{4}\right)$$

$$c) \ln(x) + x^2 = 3$$

$$f(x) = \ln(x) + x^2 - 3$$

our interval

$$(1, 2)$$

$$\text{mid point} = \frac{1+2}{2} = \frac{3}{2}$$

$$\Rightarrow f\left(\frac{3}{2}\right) = \ln\left(\frac{3}{2}\right) + \left(\frac{3}{2}\right)^2 - 3$$

$$\Rightarrow f\left(\frac{3}{2}\right) \approx 0.4055 \quad \text{root is in } \left(\frac{3}{2}, 2\right)$$

$$\text{mid point 2} = \frac{\frac{3}{2} + 2}{2} = \frac{7}{4}$$

$$\Rightarrow f\left(\frac{7}{4}\right) = \ln\left(\frac{7}{4}\right) + \left(\frac{7}{4}\right)^2 - 3$$

$$\Rightarrow f\left(\frac{7}{4}\right) \approx 0.5596 \quad \text{root is in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

$$\text{length of interval: } \frac{7}{4} - \frac{3}{2} = \frac{1}{4}$$

$$\text{mid point} = \frac{\frac{3}{2} + \frac{7}{4}}{2} = \frac{13}{8}$$

$$x = \frac{13}{8}, \text{ root in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

EX. 1.1.8, Langou

a) derive formula that relates to a, b, k and tol

After k bisections the interval $a-b$ has been halved k times so the length of final interval is:

$$\frac{b-a}{2^k}$$

If we take the midpoint of this final interval as our approximation, the forward error is at most half the length of the final interval

$$|\text{error}| \leq \frac{1}{2} \cdot \frac{b-a}{2^k} = \frac{b-a}{2^{k+1}}$$

to guarantee the forward error is within tol we need

$$\boxed{\frac{b-a}{2^{k+1}} \leq \text{tol}}$$

b) solve for k as a function of a, b , and tol .

$$\frac{b-a}{2^{k+1}} \leq \text{tol}$$

$$\Rightarrow 2^{k+1} \geq \frac{b-a}{\text{tol}}$$

$$\Rightarrow k+1 \geq \log_2 \left(\frac{b-a}{\text{tol}} \right)$$

$$\Rightarrow k \geq \log_2 \left(\frac{b-a}{\text{tol}} \right) - 1$$

$$k = \left\lceil \log_2 \left(\frac{b-a}{\text{tol}} \right) - 1 \right\rceil$$

c)

each iteration of the function requires one new function evaluation, the two end point evaluations are only needed once at the start and from then on one of two function values is always reused from the previous step. So if we evaluate f at a and b to start, then 1 more evaluation per iteration

$$\text{numEvals} = k + 2$$

d) $a = -3, b = 16, \text{tol} = 10^{-3}$

$$k \geq \log_2 \left(\frac{b-a}{\text{tol}} \right) - 1$$

$$\Rightarrow k \geq \log_2 \left(\frac{19}{10^{-3}} \right) - 1$$

$$\Rightarrow k \geq 14.2129 - 1$$

$$\Rightarrow k = 14$$

$$\text{numEvals} = k + 2$$

$$\Rightarrow 14 + 2 = \boxed{16}$$

$$e) \quad a = -0.2, \quad b = 2.7, \quad \text{tol} = 10^{-12}$$

$$k \geq \log_2 \left(\frac{2.9}{10^{-12}} \right) - 1$$

$$\Rightarrow k \geq 41.3905 - 1$$

$$\Rightarrow k = 41$$

$$\text{numevals} = k + 2 = 41 + 2 = \boxed{43}$$

EX. 1.2.4, solver 3

$$a) \quad g(x) = (2x - 1) / x^2, \quad r = 1$$

check fixed point:

$$g(1) = \frac{2(1) - 1}{1^2} = 1 = r \quad \checkmark$$

$$g'(x) = \frac{2 \cdot x^2 - (2x - 1)(2x)}{x^4}$$

$$\Rightarrow g'(x) = \frac{-2x + 2}{x^3}$$

$$\Rightarrow g'(1) = \frac{-2(1) + 2}{1^3} = \frac{0}{1} = 0$$

Since $g'(1) = 0 < 1$ locally convergent.

b)

$$g(x) = \cos(x) + \pi + 1 \quad r = \pi$$

$$g(\pi) = \cos(\pi) + \pi + 1 = -1 + \pi + 1 = \pi = r \quad \checkmark$$

$$g'(x) = -\sin(x)$$

$$g'(\pi) = -\sin(\pi) = 0$$

Locally convergent

c)

$$g(x) = e^{2x} - 1, \quad r = 0$$

$$g(0) = e^{2(0)} - 1 = 0 = r \quad \checkmark$$

$$\Rightarrow g'(x) = 2e^{2x}$$

$$\Rightarrow g'(0) = 2e^{2(0)} = 2$$

Divergent

EX • 1.2.14 Saver 3

$$a) \quad x \rightarrow \frac{1}{2}x + \frac{1}{x} \quad r = \sqrt{2}$$

$$\Rightarrow g(x) = \frac{1}{2}x + \frac{1}{x}$$

$$\Rightarrow g(\sqrt{2}) = \frac{1}{2} \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} = \sqrt{2} \quad \checkmark$$

$$\Rightarrow g'(x) = \frac{1}{2} - \frac{1}{x^2}$$

$$\Rightarrow g'(\sqrt{2}) = \frac{1}{2} - \frac{1}{(\sqrt{2})^2} = 0$$

converges at $\sqrt{2}$.

b)

$$x \rightarrow \frac{2}{3}x + \frac{2}{3x}$$

$$g(x) = \frac{2}{3}x + \frac{2}{3x}$$

$$g(\sqrt{2}) = \frac{2}{3}(\sqrt{2}) + \frac{2}{3(\sqrt{2})} = \frac{2}{3}\sqrt{2} + \frac{2\sqrt{2}}{6} = \frac{2}{3}\sqrt{2} + \frac{2\sqrt{2}}{3}$$

$$\Rightarrow \frac{2\sqrt{2}}{3} + \frac{\sqrt{2}}{3} = \frac{3\sqrt{2}}{3} = \sqrt{2} \quad \checkmark$$

$$g'(x) = \frac{2}{3} - \frac{2}{3x^2}$$

$$g'(\sqrt{2}) = \frac{2}{3} - \frac{2}{3(\sqrt{2})^2} = \frac{1}{3}$$

converges at $\sqrt{2}$

c)

$$x \rightarrow \frac{3}{4}x + \frac{1}{2x}$$

$$\Rightarrow g(x) = \frac{3}{4}x + \frac{1}{2x}$$

$$\Rightarrow g(x) = \frac{3}{4}(\sqrt{2}) + \frac{1}{2(\sqrt{2})} = \frac{3}{4}\sqrt{2} + \frac{\sqrt{2}}{4} = \frac{3\sqrt{2}}{4} + \frac{\sqrt{2}}{4} = \frac{4\sqrt{2}}{4} = \sqrt{2} \quad \checkmark$$

$$\Rightarrow g'(x) = \frac{3}{4} - \frac{1}{2x^2}$$

$$\Rightarrow g'(\sqrt{2}) = \frac{3}{4} - \frac{1}{2(\sqrt{2})^2}$$

$$\Rightarrow g'(\sqrt{2}) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

converges at $\sqrt{2}$

$$g'_a(\sqrt{2}) > 0 < g'_b(\sqrt{2}) = \frac{1}{3} < g'_c(\sqrt{2}) = \frac{1}{2}$$

a is the fastest followed by b then c .

EX. 1.3.8, Saver 3

$$r_\varepsilon \approx r - \varepsilon \frac{g(r)}{f'(r)}$$

a)

$$f(x) = x^n - ax^{n-1} = x^{n-1}(x-a) = 0$$

$$\Rightarrow f'(x) = nx^{n-1} - a(n-1)x^{n-2}$$

$$r=a$$

$$\Rightarrow f'(a) = na^{n-1} - a(n-1)a^{n-2}$$

$$\Rightarrow na^{n-1} - (n-1)a^{n-1} = a^{n-1}[n - (n-1)] = a^{n-1}$$

$$g(a) = a^n$$

$$\Rightarrow r_\varepsilon \approx r - \varepsilon \frac{g(r)}{f'(r)} = a - \varepsilon \frac{a^n}{a^{n-1}} = a - \varepsilon a$$

$$\Rightarrow r_\varepsilon \approx a(1 - \varepsilon)$$

b)

$$f_\varepsilon(x) = 0$$

$$\Rightarrow (1+\epsilon)x^n - ax^{n-1} = 0$$

$$\Rightarrow x^{n-1}[(1+\epsilon)x - a] = 0$$

$$\Rightarrow x = \frac{a}{1+\epsilon}$$

expand using geometric series

$$x_{\text{exact}} = \frac{a}{1+\epsilon} = a(1 - \epsilon + \epsilon^2 - \epsilon^3 + \dots) = \boxed{a - a\epsilon + a\epsilon^2 - \dots}$$

The sensitivity formula and the exact root match up through the linear term ϵ so the prediction is accurate for small ϵ but not exact.

CP. 1.4.2 solver 3

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

a) $f(x) = x^3 + x^2 - 1$ $x_0 = 1$
 $f'(x) = 3x^2 + 2x$

$$f(x_0) = 1$$

$$f'(x_0) = 5$$

$$x_1 = 1 - \frac{1}{5} = \frac{4}{5}$$

$$f(x_1) = \left(\frac{4}{5}\right)^3 + \left(\frac{4}{5}\right)^2 - 1 = \frac{19}{125}$$

$$f'(x_1) = 3\left(\frac{4}{5}\right)^2 + 2\left(\frac{4}{5}\right) = \frac{88}{25}$$

$$x_2 = \frac{4}{5} - \frac{19/125}{88/125} = \frac{333}{440}$$

$$x_1 = \frac{4}{5} = 0.8, \quad x_2 = \frac{333}{440} \approx 0.75681$$

b)

$$f(x) = x^2 + \frac{1}{x+1} - 3x$$

$$f'(x) = 2x - \frac{1}{(x+1)^2} - 3$$

$$f(x_0) = 1 + \frac{1}{2} - 3 = -\frac{3}{2}$$

$$f'(x_0) = 2 - \frac{1}{4} - 3 = -\frac{5}{4}$$

$$x_1 = 1 - \frac{-3/2}{-5/4} = -\frac{1}{5}$$

$$f(x_1) = \left(-\frac{1}{5}\right)^2 + \frac{1}{-\frac{1}{5}+1} - 3\left(-\frac{1}{5}\right) = \frac{189}{100}$$

$$f'(x_1) = 2\left(-\frac{1}{5}\right) - \frac{1}{\left[-\frac{1}{5}+1\right]^2} - 3 = -\frac{387}{40}$$

$$x_2 = -\frac{1}{5} - \frac{189/100}{-387/40} = \frac{359}{1985}$$

$$x_1 = -\frac{1}{5}, \quad x_2 = \frac{359}{1985} \approx 0.1809$$

4)

$$f(x) = 5x - 10$$

$$f'(x) = 5$$

$$f(x_0) = -5$$

$$f'(x_0) = 5$$

$$x_1 = 1 - \frac{-5}{5} = 2$$

$$f(x_1) = 0$$

$$f'(x) = 5$$

$$x_2 = 2 - \frac{0}{5} = 2$$

$$x_1 = 2, x_2 = 2$$

EX • 1.4.4, Saver 3

$$a) 32x^3 - 32x^2 - 6x + 9 =$$

$$f(x) = 32x^3 - 32x^2 - 6x + 9$$

$$f'(x) = 96x^2 - 64x - 6$$

$$f''(x) = 192x - 64 \quad x = -1/2$$

$$f(-1/2) = 32(-1/2)^3 - 32(-1/2)^2 - 6(-1/2) + 9 = 0 \quad \checkmark$$

$$f'(-1/2) = 96(-1/2)^2 - 64(-1/2) - 6 = 50 \neq 0$$

Apply theorem 1.11

$$f''(-\frac{1}{2}) = 192(-\frac{1}{2}) - 64 = -160$$

$$e_{i+1} \approx \frac{f''(\gamma)}{2f'(\gamma)} e_i^2 = \frac{-160}{2(50)} e_i^2 = -\frac{8}{5} e_i^2$$

$$e_{i+1} \approx -\frac{8}{5} e_i^2 \quad (\text{quadratic convergence})$$

$$x = 3/4$$

$$f(3/4) = 32(3/4)^3 - 32(3/4)^2 - 6(3/4) + 9 = 0 \quad \checkmark$$

$$f'(3/4) = 96(3/4)^2 - 64(3/4) - 6 = 0$$

multiplicity check

$$f''(3/4) = 192(3/4) - 64 = 80 \neq 0$$

multiplicity $m=2$ Theorem 1.12

$$e_{i+1} \approx \frac{m-1}{m} e_i = \frac{2-1}{2} e_i = \frac{1}{2} e_i$$

$$e_{i+1} \approx \frac{1}{2} e_i \quad (\text{linear convergence})$$

6)

$$f(x) = x^3 - x^2 - 5x - 3$$

$$f'(x) = 3x^2 - 2x - 5$$

$$f''(x) = 6x - 2$$

$$x = -1$$

$$f(-1) = 0 \quad \checkmark$$

$$f'(-1) = 0 \quad \checkmark$$

$$f''(-1) = -8 \neq 0$$

$$m=2$$

$$e_{i+1} \approx \frac{2-1}{2} e_i = \frac{1}{2} e_i \quad \text{linear convergence}$$

$$x = 3$$

$$f(3) = 0 \quad \checkmark$$

$$f'(3) = 16 \neq 0$$

$$f''(3) = 16$$

$$e_{i+1} \approx \frac{16}{2(16)} e_i^2$$

$$= \frac{1}{2} e_i^2 \quad \text{quadratic convergence}$$